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TP Artificial Neural Networks Report

Project:
Deep Learning for Crop Classification Using Multi-Source Satellite Data

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Introduction

Crop maps are a type of map that shows the location of where certain crops grow in a given area. These maps can be used to estimate how much food is produced, predict shortages, and improve economic decisions. They are a major part of sustainable agricultural development; however their creation is rather difficult especially if it is reliant on manual methods, as the location and growth of crops change frequently and data can be lost.

Automating the creation of crop maps can be done with remote sensing, which is the extraction of information about the Earth's surface from a distance. A primary method involves the use of deep learning to classify crops in a given region. This is a low-cost and effective approach that has become even more accessible thanks to various satellite datasets.

Crop map datasets include time-series images that capture the growth of crops throughout the season, as well as the labeled ground-truth (the type of crops). This is made possible thanks to Sentinel-2 **[1],** which is a satellite mission that provides high-spatial resolution observations of the earth's surface every few days. These images can be combined with Cropland Data Layer (CDL), a crop-map dataset that gives the exact location of crops across the United States of America each year. **[2]**

The primary deep learning architectures used in the state of the art are Convolutional Neural Networks, Recurrent Neural Networks, and Transformers. CNNs are particularly effective at capturing local features, while RNNs and Transformers are well designed for capturing temporal features. These architectures generally outperform classical machine learning models like SVM and random forest, due to their inability at automatic feature extraction. **[3]**

A characteristic of many crop map datasets used in the state of the art is that most of them do not include covariate data, such as climate variables, soil properties and topographic features **[4]**. These can potentially improve classification accuracy as it gives models more context to learn from.

The objectives of this study is to build and train the proposed MCTNet model with Sentinel-2 and CDL data taken from California and Arkansas, retrain the model on multi-sourced data such as environmental covariates and evaluate it, and finally build our own proposed architectures to be compared with the article’s model.

The report is **13 pages** long and is organized into the following three chapters:

- **Chapter 1:** Analysis, Reproduction and Implementation of the base classification model proposed by “A lightweight CNN-Transformer network for pixel-based crop mapping using time-series Sentinel-2 imagery.”
- **Chapter 2:** Integration of environmental covariates and evaluating the effects it has on model performance
- **Chapter 3:** The creation of our own proper models to be compared with the baseline and covariate-based model.

TABLE DES MATIÈRES :

Introduction..... 1
List of Figures..... 3
List of Tables..... 3
Chapter 1: Base Classification Model..... 1
1. Literature Review..... 1
1.1- MCTNet 1
1.2- Data Structure..... 2
2. Data Preparation and Preprocessing..... 2
2.1. Area of Study..... 2
2.2. Preprocessing..... 2
2.3. Data Splitting..... 3
3. Model Evaluation..... 4
3.1. Training Methods..... 4
3.2. Training Results..... 4
Chapter 1 Conclusion..... 5
Chapter 2: Integration of Environmental Covariates..... 6
1. Data Preparation and Analysis..... 6
1.1 Climate..... 6
Data analysis..... 6
1.2 Soil..... 7
Data Analysis..... 7
1.3 Topography..... 7
Data Analysis..... 7
2. Ablation Study :..... 7
2.1. Evaluation Results :..... 7
2.2. Evaluation Comparaison:..... 8
Conclusion of Adding Covariates :..... 8
Chapter 3: Model Design and Improvement..... 9
1. Proposed Model and Literature Review..... 9
1.1 InceptionTime:..... 9
1.1.1 - Literature :..... 9
1.1.2 - Architecture :..... 9
1.1.3 - Training :..... 9
1.1.4. Evaluation :..... 10
1.2 HybridCropNet :..... 10
1.2.1 - Literature :..... 10
1.2.2 - HybridCropNet Architecture..... 10
1.2.4 - Evaluation..... 11
1.3 IncepTAE-CovLF..... 12
1.3.1 - Literature :..... 12

1.3.2 - IncepTAE-CovLF Architecture....12
1.3.3 - Training....12
1.3.4 - Evaluation....13
2. Model Comparison....13
Project Conclusion :....14
REFERENCES....1
ANNEXE A....2
Diagrams of the Model Architectures....2
Hybrid Crop Net....3
IncepTAE-CovLF :....3
InceptionTime :....4

List of Figures

1. Figure 1. General architecture of the MCTNet model.
2. Figure 2. Initial class distribution in Arkansas and California.
3. Figure 3. Mean NDVI profiles by crop class and region.
4. Figure 4. Class distribution after preprocessing.
5. Figure 5. Final class distribution after grouping minority classes.
6. Figure 6. Accuracy and loss curves of MCTNet.
7. Figure 7. Confusion matrix of MCTNet.
8. Figure 8. Performance comparison after adding environmental covariates.
9. Figure 9. Accuracy and loss curves of the InceptionTime model.
10. Figure 10. Accuracy and loss curves of the IncepTAE-CovLF model.
11. Figure 11. Overall performance comparison of all implemented models.
12. Figure A.1. Architecture of the HybridCropNet model.
13. Figure A.2. Architecture of the IncepTAE-CovLF model.
14. Figure A.3. Architecture of the InceptionTime model.

List of Tables

1. Table 1. Data split between training, validation, and test sets.
2. Table 2. Evaluation results of the MCTNet model.
3. Table 3. Comparaison between our MCTNet model and the article.
4. Table 4. Average climate statistics by state.
5. Table 5. Soil property statistics by state.
6. Table 6. Topographic statistics by state.
7. Table 7. Ablation study results with environmental covariates.
8. Table 8. Performance comparison of the InceptionTime model.
9. Table 9. Performance comparison of the HybridCropNet model.
10. Table 10. Performance comparison of the IncepTAE-CovLF model.

Chapter 1: Base Classification Model

1. Literature Review

The article “A lightweight CNN-Transformer network for pixel-based crop mapping using time-series Sentinel-2 imagery.” **[5]** seeks to combine the strength of Convolutional Neural Networks and Transformers for crop-map classification without requirement of extensive data-labeling. Missing values caused by clouds or lack of observation have a large impact against these proposed models, therefore they proposed an enhanced architecture known as MCTNet.

1.1- MCTNet

The Multi-stage CNN-Transformer Network (MCTNet) model is a hybrid deep learning architecture that combines Convolutional Neural Networks and Transformers to capture both local spatial patterns and long-term temporal features in crop growth across the seasons. It is structured into the following layers:

- **Input:** time-series spectral images and their changes over a 360-day period (10 × 36)
- **CTFusion:** The model has three layers called “stages”, it is the main fusion block of a CNN module and a Transformer sub-module, where their outputs are concatenated and processed with a max pooling operation.
- **CNN Sub-module:** Simple CNN block that consists of two 1D convolution layers, each followed by batch normalization layers. The output is then passed into a ReLU activation function to introduce non-linearity.
- **Transformer Sub-module:** Uses the same encoder as the original Transformer architecture with a designed attention-based learnable positional encoding (ALPE) module, which helps with understanding which temporal data is unreliable (has clouds and missing values).
- **MLP Layer:** Final classification layer that predicts the crop type.

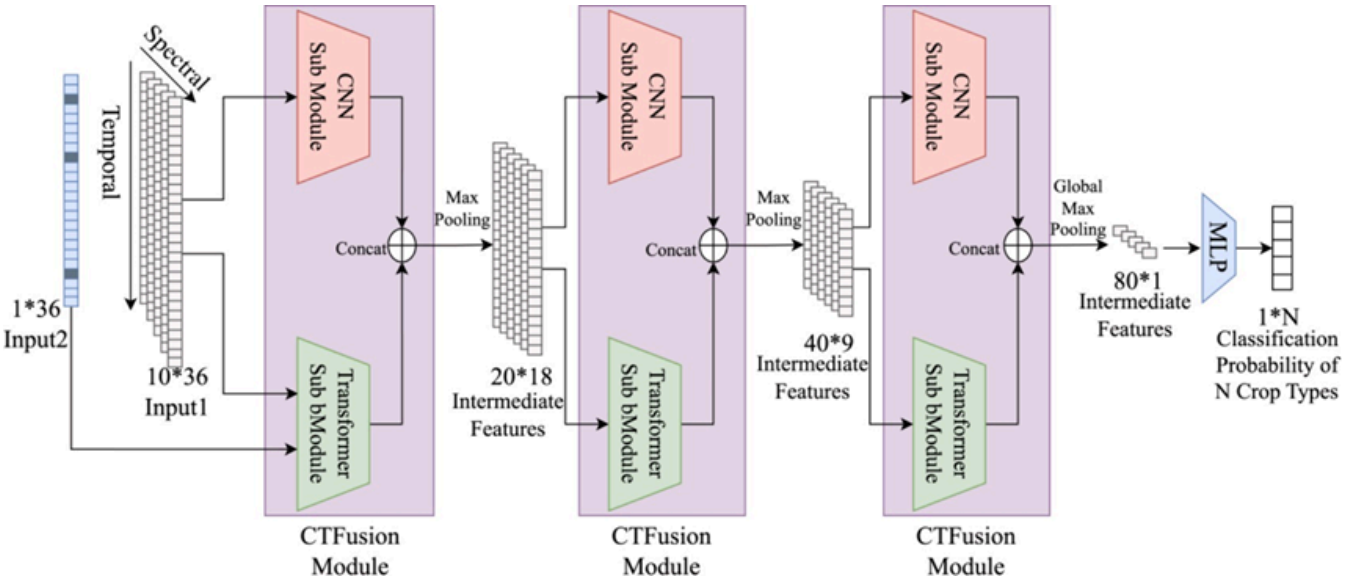


Figure 1. General architecture of the MCTNet model.

1.2- Data Structure

The article uses Sentinel-2 Level-2A spatial imagery and the USDA Cropland Data Layer on two states : Arkansas and California, with two mapping areas randomly selected per state. A total of 10,000 labeled samples were collected, each consisting of 10 spectral bands across 36 temporal observations, which are spatial imagery of 10 days across the year of 2021.

For preprocessing they removed cloud-affected pixels and replaced null values with zeros, and in order to mask non-cropland areas they used ESA WorldCover 2021 data. They also grouped crop types that constitute less than 5% of the total number of samples into the category “others”.

2. Data Preparation and Preprocessing

2.1. Area of Study

In accordance with the article, we chose two random areas in Arkansas and California. Before preprocessing, we noticed that Arkansas is composed of large fields and diverse crops such as rice, soybeans, corn, and cotton, while California contains mainly rice, and then grapes, tree crops, citrus, cherries and unused cropland.

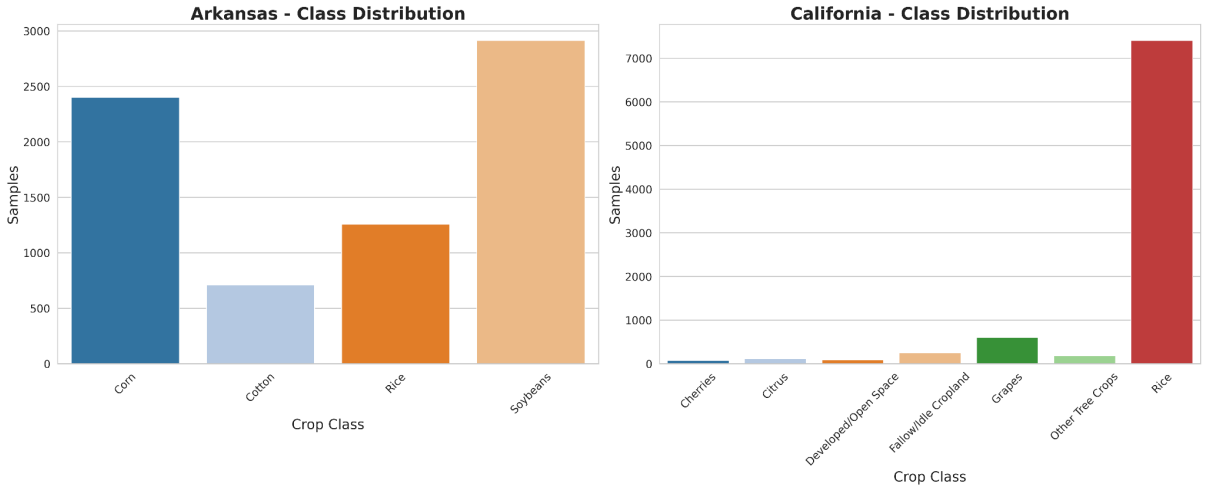


Figure 2. Initial class distribution in Arkansas and California.

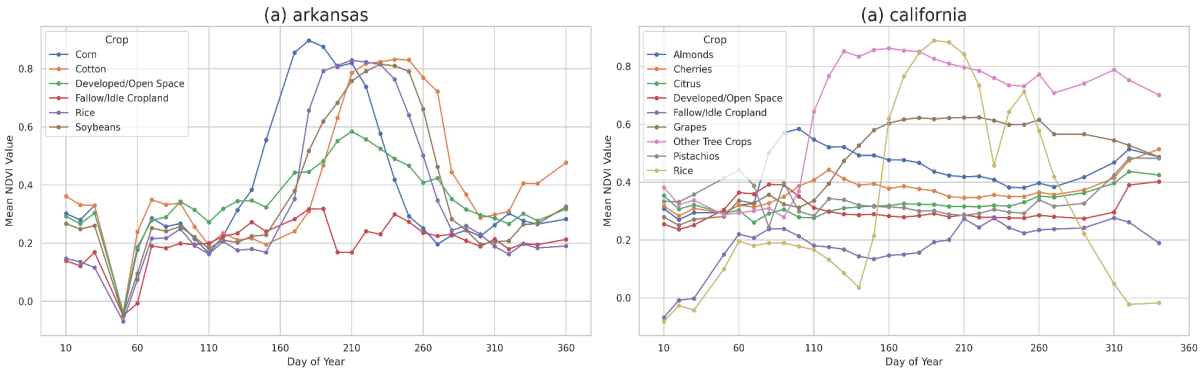


Figure 3. Mean NDVI profiles by crop class and region.

2.2. Preprocessing

The areas were selected and exported using **Google Earth Engine [6]**. We took Sentinel-2 Layer-2A data of 2021 and extracted the time-series data we need for the model, which is 10 bands with 36 time-steps (each 10-bands relate to spectral images of 10 days across a year). Cropland Data Layer was combined to obtain the exact location of crops as well as their classification. Then we followed the exact preprocessing methods used in the paper, we removed the cloud-covering, replaced the null

values with zeros, and used ESA WorldCover 2021 **[7]** to mask non-cropland areas. The final output is a .csv file for each four sub-regions: arkansas1, arkansas2, california1, and california2.

Since the regions we chose aren't the exact same as the article, the resulting sample size was **16381** amongst them was rice occupying the majority of the samples. We reduced the amount of rice by 5000, bringing our final sample count to **11381** samples.

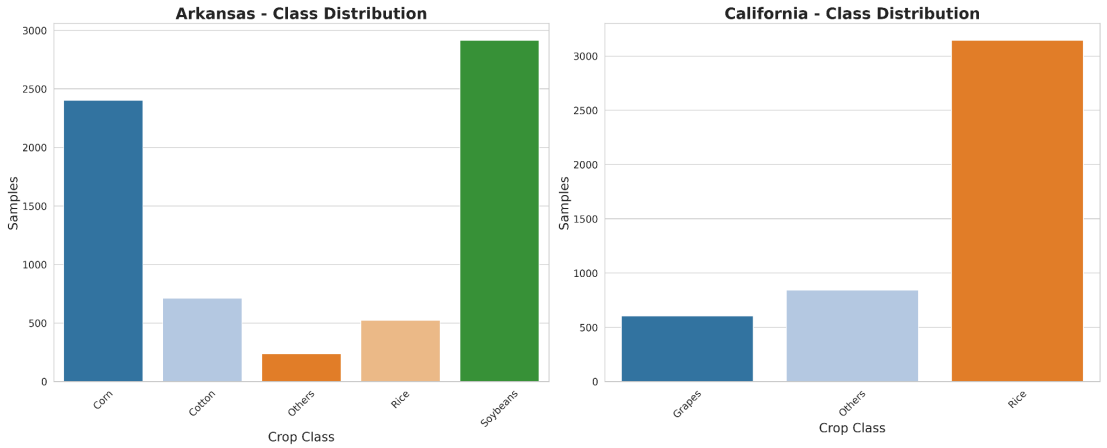


Figure 4. Class distribution after preprocessing.

We noticed that some classes were under-represented, therefore we grouped crop types that constitute less than 2% of the total number of samples into the category “others” **[figure 5]** which helped include grapes and cotton as their own class. Bringing the total number of classes into **6 classes**.

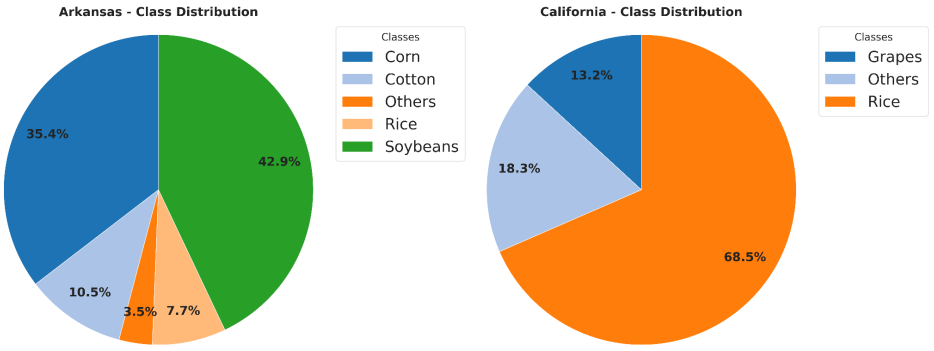


Figure 5. Final class distribution after grouping minority classes

2.3. Data Splitting

Data was split using the same distribution as the article, samples are randomly selected with a rate of 8:2 for training and validation respectively, with most of the samples used for testing. The samples for each data is shown in the below table.

Sous-ensemble	Forme
Train	(1916, 36, 10)
Validation	(420, 36, 10)
Test	(9045, 36, 10)

Table 1. Data split between training, validation, and test sets.

3. Model Evaluation

3.1. Training Methods

The MCTNet model is trained with the **Adam** optimizer, with the learning rate set to **0.001**. Since this is a multi-class classification problem, the loss function used is **Sparse Categorical Crossentropy**. Training is done over **20** epochs with a batch size of **32**.

Call backs such as EarlyStopping and ModelCheckpoint are used to get the most accurate model based on the lowest validation loss, EarlyStopping stops the training if the validation loss keeps increasing after **5** epochs while ModelCheckpoint saves the model with the best validation loss.

Performance is calculated with metrics such as :

- **Overall Accuracy (OA) :** Percentage of correctly classified samples out of total samples.
- **F1-score :** Mean of precision and recall, balances false positives and false negatives.
- **Kappa coefficient :** Measures how much the model is better than random guessing

3.2. Training Results

From the accuracy graph we notice that the model immediately gets high performance on training data, and slowly learns to generalize on validation data. Validation accuracy starts to stabilize closer to training accuracy at epoch 8. From the loss graph, we notice similar behavior where training loss is immediate and validation loss slowly approaches 0. Since the model is generalizing for validation data, then there is no sign of overfitting and the model is training correctly.

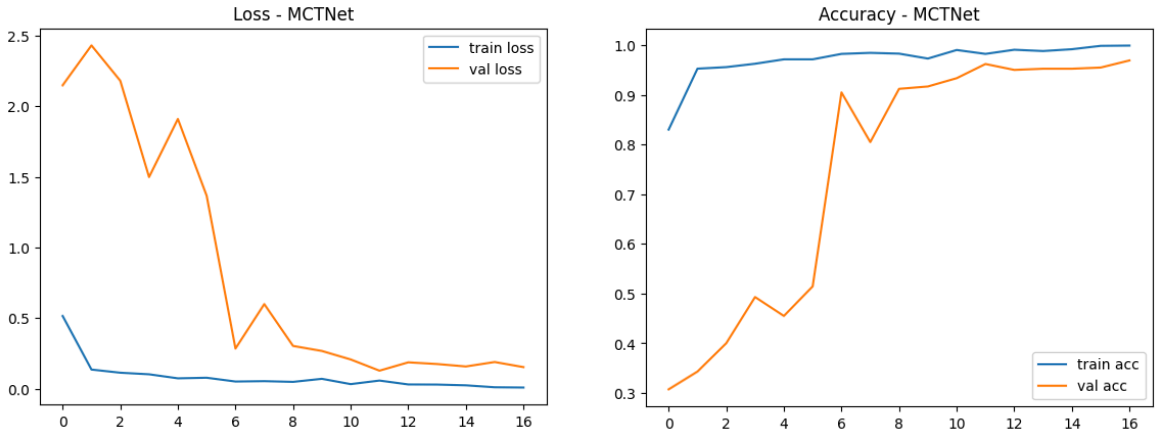


Figure 6. Accuracy and loss curves of MCTNet

The final evaluation metrics are as follows :

Metric	Score
OA	**0.977**
Kappa	**0.969**
F1-Score	**0,940**

Table 2. Evaluation results of the MCTNet model.

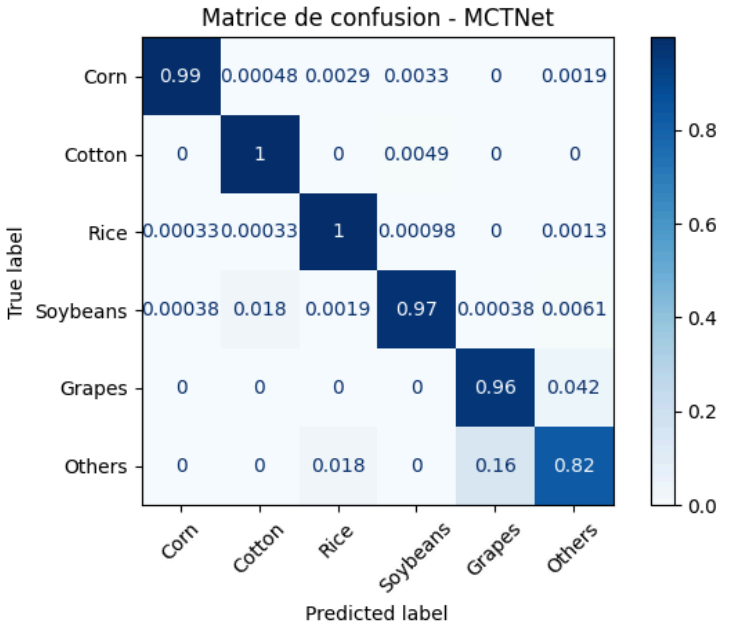


Figure 7. confusion matrice of MCTNet.

From the confusion matrix, we notice a concentration around the diagonal which means the model is correctly predicting each class. The classes with the lowest samples include lower validation accuracy (grapes with 0.96 and others with 0.82), which is logical since grapes have low samples and others contain data with variations. This can be observed as a small sign of class imbalance.

Model	OA	Kappa	F1-Score
Trained MCTNet	**0.977**	**0.969**	**0.940**
Article's MCTNet	0.91	0.8785	0.881

Table 3. Comparaison between our MCTNet model and the article.

Chapter 1 Conclusion

In this chapter, we reviewed the methods of the article and defined the architecture of MCTNet. Sentinel-2 and CDL were used as the main dataset with 2 areas chosen in the state of Arkansas and California. The Sentinel-2 data got reshaped into 36 temporal steps and 10 bands and finally organized into six crop classes using CDL. The final trained model achieves an OA score of 97.7%, higher than the article's results.

Chapter 2: Integration of Environmental Covariates

1. Data Preparation and Analysis

The environmental covariates were acquired through Google Earth Engine, then projected to align with our dataset. These three datasets were used:

- ERA5-Land Daily Aggregated for climate **[8]**
- OpenLandMap for soil **[9][10]**
- USGS SRTMGL1_003 for topography. **[11]**

The final export is a .csv file that combines Sentinel-2 bands, environmental covariates, the crop labels, and the region.

1.1 Climate

Climate data were extracted from the ERA5-Land Daily Aggregated product, This climate data is organised into 5 variables captured every day for the year 2021

- temperature_2m temperature_2m_max
- temperature_2m_ min total_precipitation_sum
- u_component_of_wind_10m v_component_of_wind_10m.

These variables represent respectively: average temperature, maximum and minimum temperature, daily accumulated precipitation, the east-west and north-south wind components above 10 meters.

To make it consistent with our existing sentinel-2 time-series, we took 36 climate data averaged through 10 days and organized them to fit the 36 temporal observations. Each time step combines 10 sentinel-2 bands and 6 climate variables, thus giving us a shape of (36, 16).

Data analysis

State	Temperature (C)	Min temp (C)	Max temp (C)	Precipitation (mm)	Wind (m/s)
Arkansas	17.12	-10.28	33.05	3.7	1.22
California	19.14	3.59	41.70	1.6	1.18

Table 4. Average climate statistics by state.

From the minimum and maximum temperatures, we notice that Arkansas is a colder region than California, while California is drier and less humid than Arkansas due to its much lower precipitation. Crops develop differently based on temperature and humidity and thus can look different, this type of data can help the model distinguish between crops by providing more context.

1.2 Soil

The soil covariates were obtained from OpenLandMap. Two non-temporal variables are provided : soil_oc and soil_ph, corresponding to soil organic carbon and soil pH. Since they are not temporal they were integrated directly into our dataset.

Data Analysis

State	Soil pH	Soil Organic Carbon
Arkansas	5.8	2.37
California	7.27	3.43

Table 5. Soil property statistics by state.

We notice an obvious difference between both soil pH and organic carbon for Arkansas and California, Arkansas has more acidic soil while California has higher organic content. Crop growth is highly dependent on soil nature, this will provide very important insight on which crops grow under which soil.

1.3 Topography

The topographic variables are taken from USGS SRTMGL1_003, a digital elevation model with 30 m spatial resolution. Three non-temporal variables are used: topo_elev, topo_slope, and topo_aspect, corresponding to elevation, slope, and aspect.

Data Analysis

State	Elevation	Slope
Arkansas	61.40	2.08
California	52.85	2.10

Table 6. Topographic statistics by state.

We notice that Arkansas is slightly higher in elevation compared to california indicating taller mountainous land, however slope is roughly equivalent which means they have similar mountain ranges. This indicates that elevation and slope may not differentiate much between arkansas and california, however will still provide useful context on the type of crops that grow in certain elevations for the model.

2. Ablation Study :

2.1. Evaluation Results :

To better understand the impact of the added data, we trained four models each with different inputs of covariates. The comparison is between the baseline model (the model trained in the previous chapter), model trained with climate data, model trained with soil data, model trained with topographic data, and finally the model trained with all covariates. Below is a comparative table of the results according to the metrics used in the MCTNet article.

Configuration	OA	Kappa	F1
Baseline (Sentinel)	0.977	0.969	0.940
Sentinel + Climate	0.985	0.981	0.976
Sentinel + Soil	0.985	0.981	0.976

Sentinel + Topo	0.981	0.976	0.968
Sentinel + All	0.983	0.978	0.973

Table 7. Ablation study results with environmental covariates.

2.2. Evaluation Comparaison:

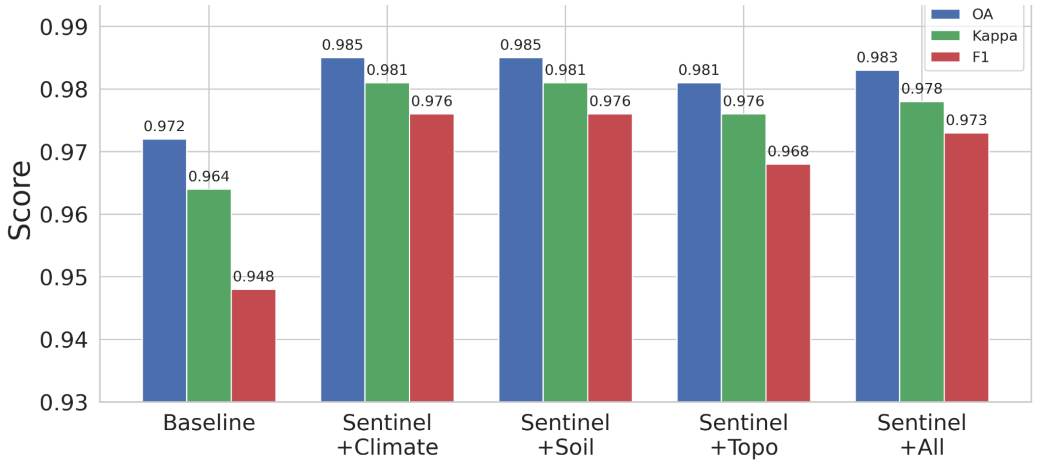


Figure 8. Performance comparison after adding environmental covariates.

The baseline model follows the same evaluation metrics as the previous chapter. We notice that when **climate data** is added, performance improves significantly (OA = 0.985, F1 = 0.976). This seems logical, as climate plays a direct role in crop growth and therefore provides relevant additional information.

The same observation applies to **soil data**, which shows exactly the same level of improvement. This confirms that soil characteristics are also very important for distinguishing the different classes.

In contrast, adding **topographic data** provides a more limited benefit. Performance increases, but remains slightly below that achieved with climate or soil data alone. This suggests that topography has a more indirect impact.

Finally, when all the data are combined, the results lower slightly as compared to climate and soil-only data. This can be explained by the fact that some information becomes redundant or introduces noise, which complicates the model's work.

Conclusion of Adding Covariates :

In this chapter, we conclude that adding covariates data **can improve performance**, but only if that data provides genuinely useful information. Adding all available sources is therefore not always the best strategy. In general covariates can improve model metrics, the most useful of which are climate and soil data, with topography also showing slight improvement.

Chapter 3: Model Design and Improvement

1. Proposed Model and Literature Review

We propose three different models intended for three different uses, all of them are implemented for the intention to research various time-series models that can challenge MCTNet while also providing simpler architectures. All models are trained on the **baseline data** (non-covariates) from chapter 1.

1.1 InceptionTime:

MCTNet is a transformer-CNN hybrid with a complex architecture, our main interest for our first model is to look over the results of applying **CNNs only** for time-series data. CNNs are much faster to train and are more adept at local feature extraction. The main concept we fixated on is the Inception architecture, one of the most popular CNN architectures.

1.1.1 - Literature :

Inception is a model introduced by Google in the article *"Going Deeper with Convolutions"* **[12]**, the unique aspect of it is the use of the Inception modules: A module that contains several parallel convolutional layers each with their own kernels, which are concatenated into a single output. This model is precise in image data, but cannot properly treat temporal data. InceptionTime is the temporal version of this architecture, first introduced in the article *"InceptionTime: Finding AlexNet for Time Series Classification"* **[13]** , it uses 1D CNNs with larger kernels intended to capture temporal patterns that are more spread-out. This model is implemented for crop classification in the article *"Machine Learning-Based Summer Crops Mapping Using Sentinel-1 and Sentinel-2 Images"* **[14]**, where it achieves promising results.

1.1.2 - Architecture :

The model is made out of 6 inception modules (called depth), they all consists of the three following layers :

- **Parallel Conv1D (kernel sizes 9, 19, 39):** The main part of each Inception module. Three convolutions run on the same input with kernel sizes 9, 19, and 39
- **Bottleneck Conv1D (kernel = 1):** Compresses the channel dimension before each parallel convolution.
- **MaxPooling1D branch:** Pooling layer that comes right after each parallel convolution.

The remaining layers are applied after each inception modules and include the following :

- **Residual connections:** prevents vanishing gradients
- **BatchNormalization:** Applied after regularization mechanism, replacing dropout entirely.
- **GlobalAveragePooling1D:** sends the temporal dimension into a fixed size, ready to be classified
- **Dense Layer :** Classification layer for the 6 classes with softmax as activation function.

1.1.3 - Training :

The model is trained for 40 epochs due to slower generalization at earlier epochs. ReduceLROnPlateau is also used to maximize the convergence to the solution without manually setting the learning rate. Similar to MCTnet, its initial learning rate is 0.0001 and uses the Adam optimizer. From the graphs we notice a stable training with no sign of overfitting.

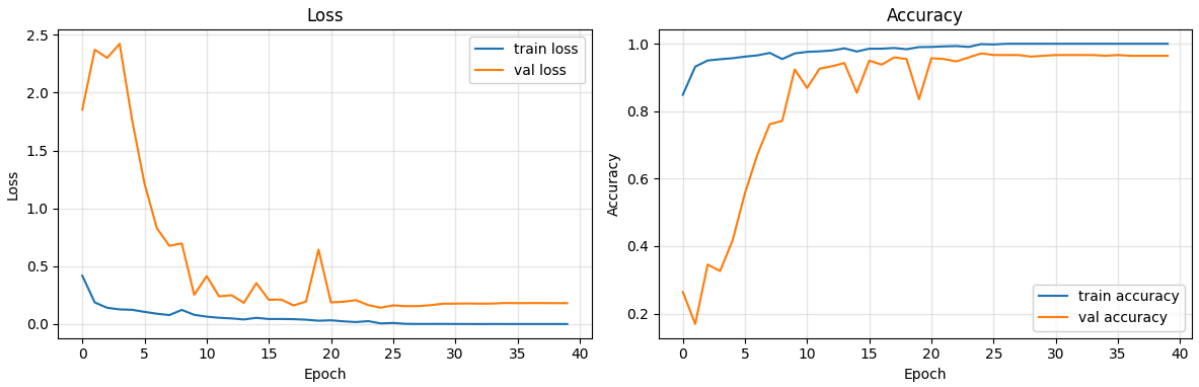


Figure 9. Accuracy and loss curves of the InceptionTime model.

1.1.4. Evaluation :

Model achieves the following results :

Model	OA	Kappa	Macro F1
MCTNet (baseline)	0.977	0.969	0.940
MCTNet (all covariates)	0.983	0.978	0.973
InceptionTime	**0.983**	**0.977**	**0.960**

Table 8. Performance comparison of the InceptionTime model.

We notice that InceptionTime achieves results superior to the baseline MCTNet, with results closer to the all-covariates trained MCTNet model.

1.2 HybridCropNet :

1.2.1 - Literature :

CBAM (Convolutional Block Attention Module) is an attention mechanism that is widely used for improving CNNs **[16]**. It directs the architecture into the important features for extraction. This attention module was used in the article "A new attention-based CNN approach for crop mapping using time series Sentinel-2 images" **[17]** in which it was adopted for geographic data showing superior results. HybridCropNet is a proposed model of ours that aims to include CBAM as well as the use of both CNNs and Transformers and incorporation of optimization techniques such as depthwise convolutions, a type of convolution used in MobileNet **[18]**.

1.2.2 - HybridCropNet Architecture

The model takes as input a time series (Sentinel-2 images), a mask to handle missing data, and static data (example: the region). Static data is integrated using a cross-attention mechanism, which allows predictions to be adapted to the context (such as the region or soil characteristics). The model uses these 3 main parts :

- **CBAM :** allows for better information capture. Unlike ECA used in MCTNet, CBAM takes into account both the importance of variables and their evolution over time, making it more suitable for time series.
- **Gated Fusion :** Inspired by RNN models like GRU and LSTM, it allows for intelligent information combination. Instead of a fixed fusion, the model learns to adjust the importance of the CNN and Transformer outputs based on the data.
- **Optimisation Layers :** The model is optimized with several efficient techniques: depthwise convolutions, regularization methods like DropPath, and pre-norm.

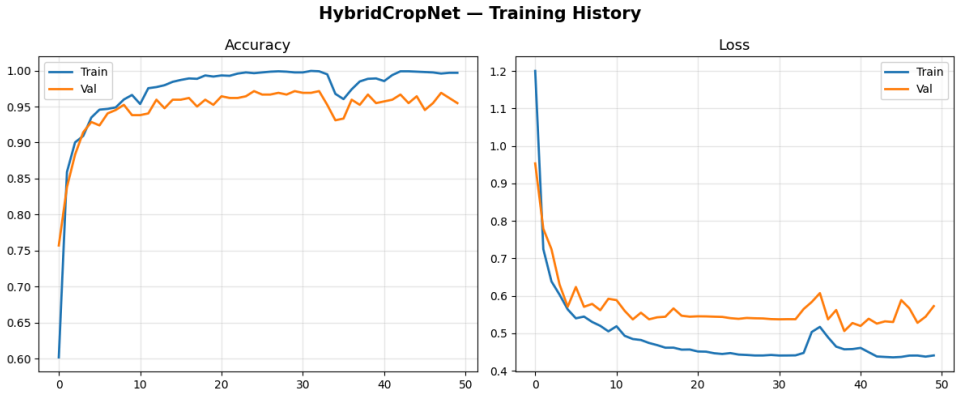
The model incorporates a set of layers called GatedCTFusion, each block integrates :

- Adaptive Local Positional Encoding ALPE (a type of enhanced positional encoding)
- CNN layer for local feature extraction
- Transformer layer for global relation extraction
- Attention Mechanism CBAM,
- An adaptive fusion (gated fusion)

To produce the final result, the model combines several summaries of the time series (maximum, average and last observation), then performs the classification. A diagram showing the global architecture of the model is provided in **[Annexe A]**

1.2.3 – Training :

HybridCropNet was trained for up to 80 epochs using the **AdamW** optimizer with a weight decay of 0.00001, a cosine decay schedule, and a starting learning rate of 0.001. To avoid keeping a suboptimal model, early stopping and checkpointing were applied based on the validation F1-macro score, which assures that only the best weights are saved. To address the obvious existence of class imbalance, label smoothing and class weights got included into the loss function. From the graphs we notice that the training goes smoothly without a sign of overfitting.



1.2.4 - Evaluation

Model achieves the following results :

Model	OA	Kappa	Macro F1
MCTNet (baseline)	0.977	0.969	0.940
MCTNet (all covariates)	0.983	0.978	0.973
HybridCropNet	0.978	0.971	0.954

Table 9. Performance comparison of the HybridCropNet model.

1.3 IncepTAE-CovLF

1.3.1 - Literature :

Our main objective in the 3rd model is an architecture that is **lightweight**. The article '*Satellite Image Time-Series Classification with Inception-Enhanced Temporal Attention Encode*r' **[15]** was reviewed as the main reference, it follows the lightweight IncepTAE model which provides the most relevant trade-off between predictive performance and computational cost. This backbone got extended through late fusion of static covariates, resulting in our custom model, IncepTAE-CovLF (CovLF stands for Covariates Late Fusion).

1.3.2 - IncepTAE-CovLF Architecture

The proposed architecture combines a temporal branch, responsible for learning crop phenological dynamics, and a static branch, dedicated to environmental context. Their late fusion enables the effective use of covariates while preserving a lightweight model. A diagram showing the summary of this architecture is included in **[Annexe A]**

- **Embedding and Positional Encoding :** This stage preserves the temporal order of observations and prepares the sequences for the attention mechanism. It performs normalization of the temporal series, projection of the inputs into a latent space, and addition of positional encoding based on acquisition dates.
- **Inception-Enhanced Module :** This module extracts more discriminative temporal patterns, it performs enhancement of the temporal representation through convolution and pooling operations and combination of global and local information.
- **Self-Attention Module :** This module models long-range temporal dependencies and produces computation of the Q, K, and V matrices, generation of an attention map through dot-product and softmax operations, weighted aggregation of values.
- **Static Covariate Branch :** This branch integrates static environmental context by separating the processing of soil and topographic variables, encoding through a lightweight MLP and the projection of the 5-dimensional static vector into a 16-dimensional latent representation.
- **Late Fusion** : This stage combines temporal information and static covariates, it performs the concatenation of the temporal representation and static representation, projection through an intermediate dense layer, and the final 6-class softmax classification.

1.3.3 - Training

The model is trained for a maximum of 40 epochs. It uses similar callbacks as the previous models, Adam optimizer and a learning rate of 0.0001. From the graph we notice correct training with no sign of overfitting.

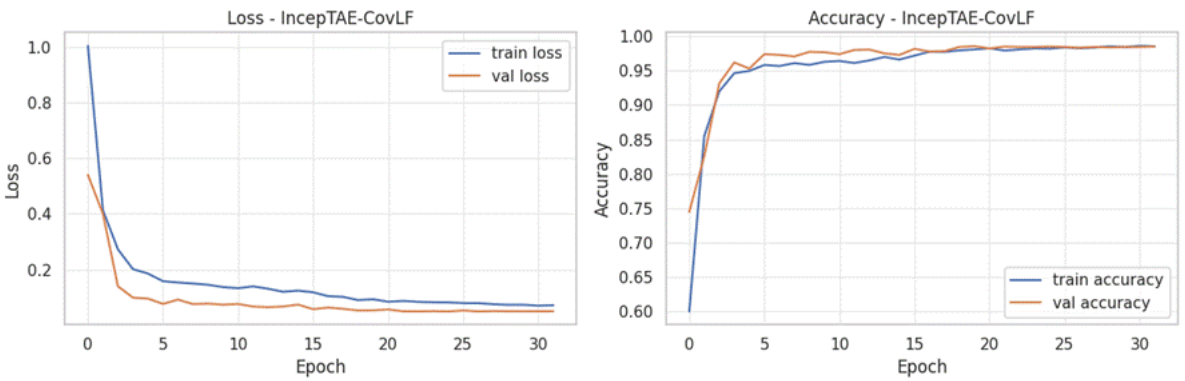


Figure 10. Accuracy and loss curves of the IncepTAE-CovLF model.

1.3.4 - Evaluation

Model achieves the following results :

Model	OA	Kappa	Macro F1
MCTNet (baseline)	0.977	0.969	0.940
MCTNet (all covariates)	0.983	0.978	0.973
IncepTAE-CovLF	0.982	0.977	0.966

Table 10. Performance comparison of the IncepTAE-CovLF model.

2. Model Comparison

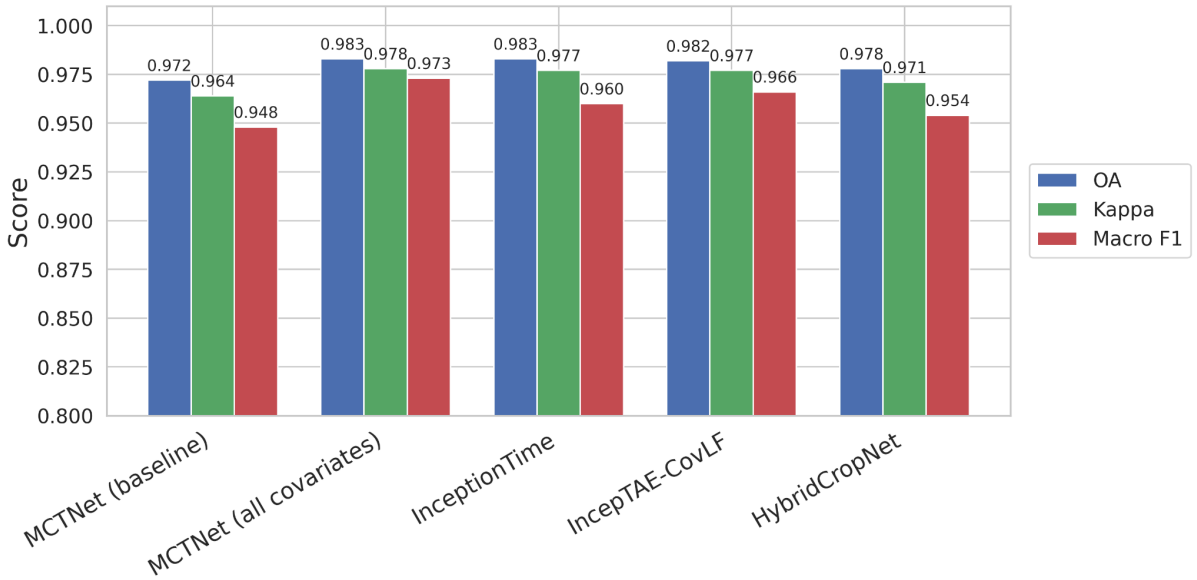


Figure 11. Overall performance comparison of all implemented models.

We notice that despite being a purely CNN-based model with no attention mechanism, InceptionTime gets an increased OA score at **0.983** and Kappa at **0.977** closer to the covariates-based MCTNet model. This result demonstrates that convolutions alone can be comparable with more complex architectures.

The HybridCropNet model achieves an OA score of **0.978** which is an overall improvement compared to the baseline model.

The IncepTAE-CovLF model also achieves a good performance comparable to the covariates-model with an OA score of **0.982**, despite being the most lightweight model in terms of parameters. This suggests that the addition of static covariates with late fusion can provide benefits.

This chapter suggests that the implementation of different architectures can yield results similar to the covariate-enhanced model, showing the importance of choosing the right deep learning architecture.

Project Conclusion :

In this project, we prepared the Sentinel-2 and CDL data by choosing 2 random locations in both Arkansas and California in the year of 2021. We implemented the article's proposed model MCTNet and trained it on time-series data, achieving an OA score of **0.977**. We also studied the use of environmental covariates such as climate, soil and topographic data in which we noticed that their inclusion did indeed improve the performance of the model, achieving an OA score of **0.983** if all covariates are included. Finally, we developed 3 models of our own each in which the best model achieved an OA score of **0.983**.

Perspectives :

For future work, here is our perspectives :

- The areas chosen for california and arkansas are small in size and lack crop variation, no method for resolving class-imbalance is used.
- More states with different environmental covariates should be tested.
- Increase the samples of our data by choosing more than the year 2021, for example we could take 2-3 years worth of data. This would give us models more capable of generalizing on never-seen data.
- Enhancing our proposed models with environmental covariates and evaluating their results.

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ANNEXE A
Diagrams of the Model Architectures

Hybrid Crop Net

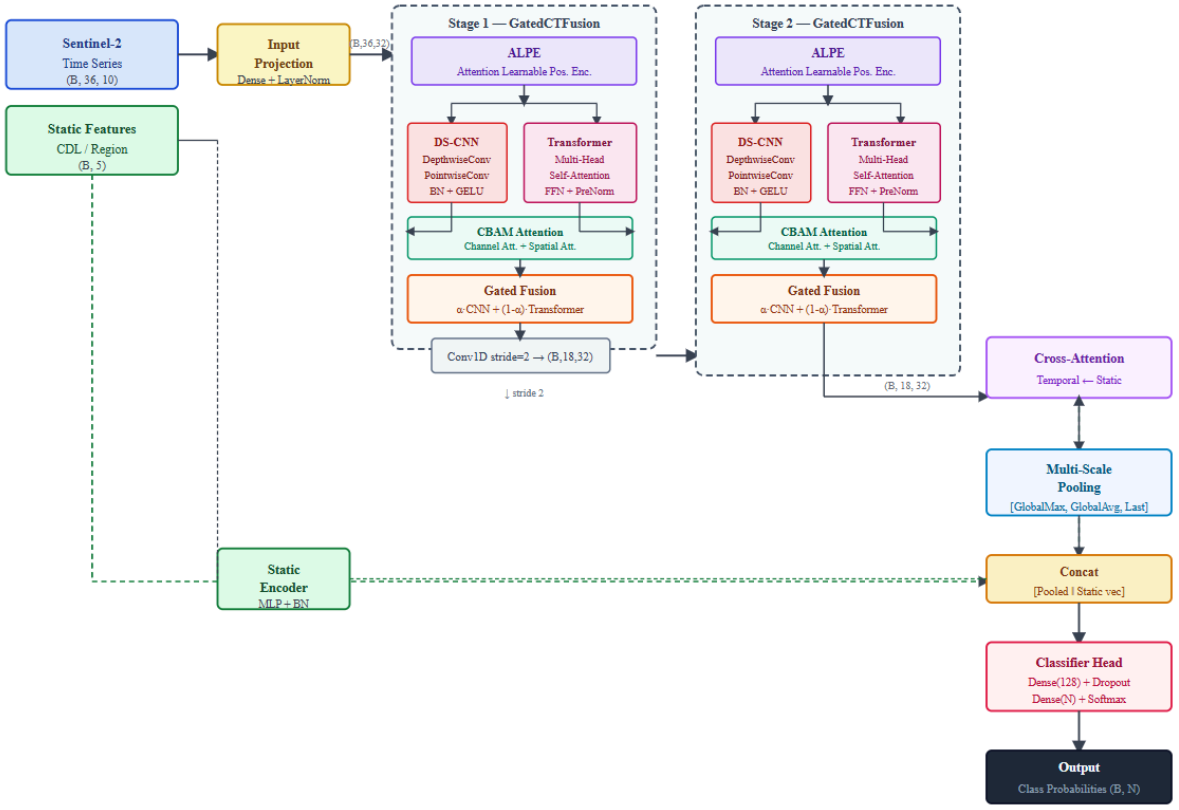


Figure A.1. Architecture of the HybridCropNet model.

IncepTAE-CovLF :

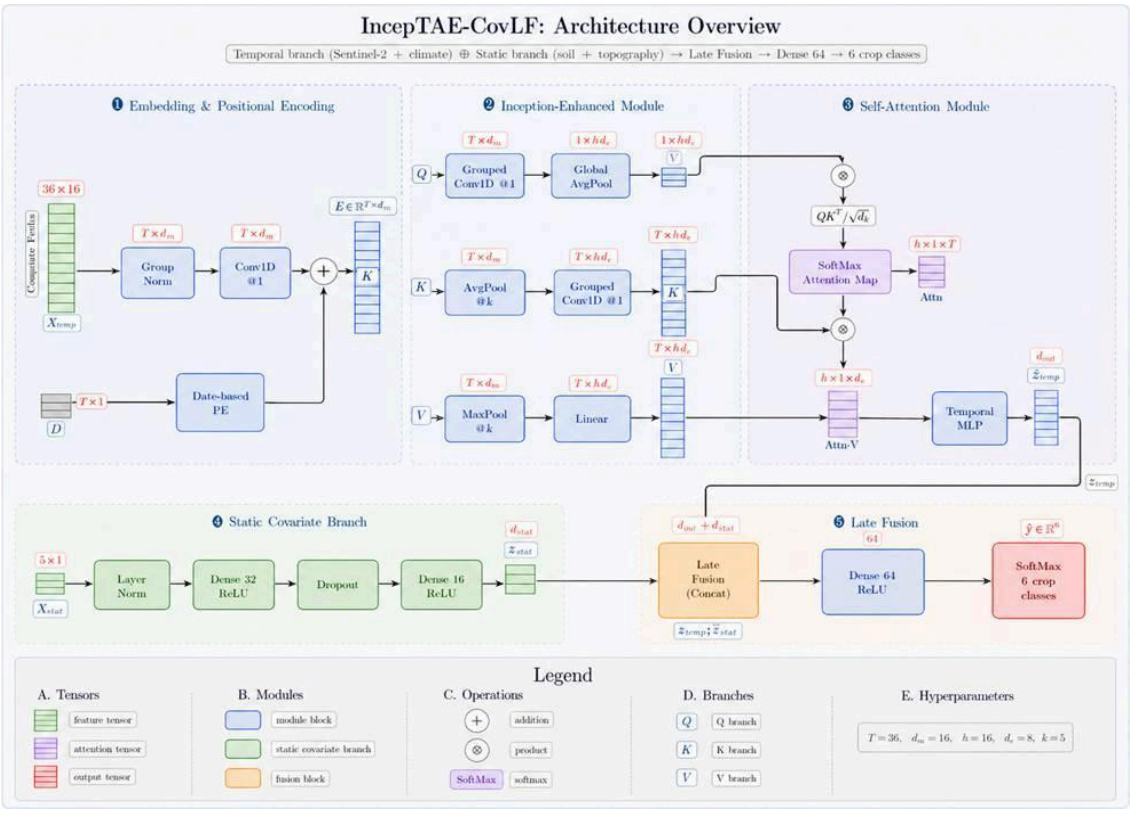


Figure A.2. Architecture of the IncepTAE-CovLF model.

InceptionTime :

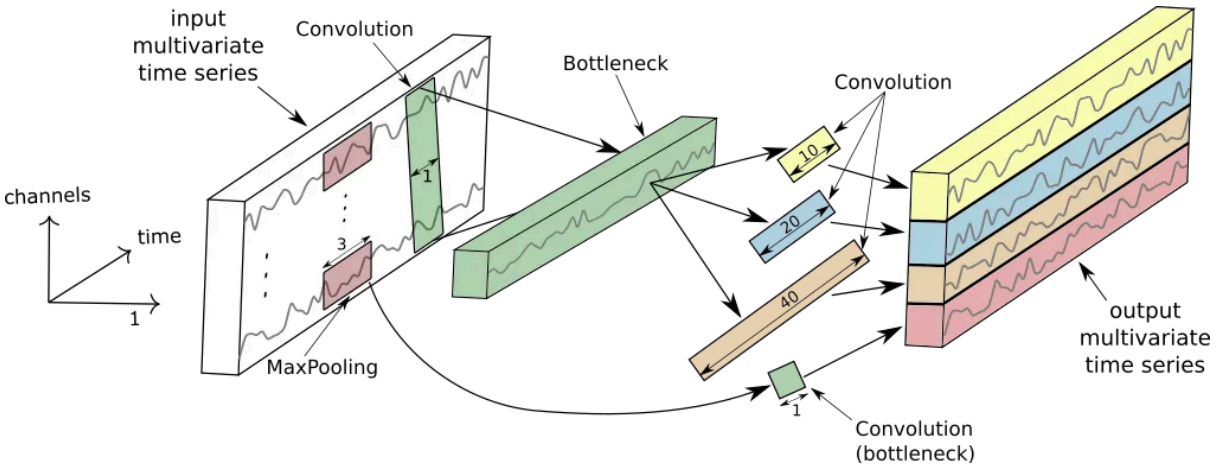


Figure A.3. Architecture of the InceptionTime model.